Law of conservation of mass

Purpose

To attempt to verify & observe the Law of Conservation of Mass - In any chemical reaction, the total mass of the reactants is always equal to the mass of the products.

You will need

Balance (accurate to 0.01 g), 100mL beaker, 50mL vinegar, 10g baking soda (sodium hydrogen carbonate), 100mL conical flask, balloon, weighing paper, graduated measuring cylinder, funnel.

What to do

1. Record the mass of the empty 100mL beaker. Pour about 15 mL of vinegar into the beaker. Record the mass of the beaker & vinegar.
2. Weigh out ~5 g of baking soda onto a piece of weighing paper. Record the exact mass of baking soda used.
3. Place the beaker/vinegar and the weighing paper/baking soda on the pan of the laboratory balance. Measure and record the mass of these reactants/containers.
4. Carefully pour the baking soda into the beaker of vinegar and allow the reaction (fizzing and bubbling) to complete. Measure and record the mass of the remaining products.

5. Record the mass of the empty conical flask. Pour about 15 mL of vinegar into the 100 mL flask using the funnel to make certain the neck of the flask does not get wet. Obtain the mass of the flask & vinegar.
6. Prepare the sample of baking soda as before and now determine and record the mass of the reactants, the flask and the balloon (which will be the cover for this container)
7. Add the baking soda to the flask and quickly cover the mouth of the flask with the balloon.
8. Measure and record the mass of the products/container/balloon.

Results: trial one

Item	Mass (grams)
Beaker + Vinegar	
Baking soda	
Total (beaker and reactants)	
Final mass (after reaction)	
Difference	

Results: trial two

Item	Mass (grams)
Conical flask + Vinegar	
Balloon	
Baking soda	
Total (flask, balloon and reactants)	
Final mass (after reaction)	
Difference	

Questions

What evidence was there that a chemical reaction occurred?

Account for the difference in mass for trial one.

Account for the difference in mass for trial two.

Did the experiment demonstrate the law of conservation of mass? Explain.

Suggest two improvements that could be made to the method that would increase the accuracy of this demonstration.